STAT6611

- Lecture 12 (R, Oct 1)
- solutions posted to Homework 1
- Homework 2 assigned due F, Oct 2 by midnight (this week)
  - practice on LSE (without constraints)
- Homework 3 assigned due F, Oct 16 by midnight

- The midterm will be some type of take-home exam about 3 weeks from now (more details next time)

Corollary 1. Under Model III, $c+\underline{a}'\underline{y}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if $H\underline{a} \in \mathcal{C}(X)$ and $E(c+\underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$.

iff $c=0$ & $\underline{q}$ is part of solution to $\begin{pmatrix} H & X \\ X' & 0 \end{pmatrix}\begin{pmatrix} \underline{q} \\ \underline{p} \end{pmatrix} = \begin{pmatrix} \underline{0} \\ \underline{\lambda} \end{pmatrix}$ for some $\underline{p}$

iff $c=0$ & $X'\underline{q} = \underline{\lambda}$ & $H\underline{q} + X\underline{p} = \underline{0}$ (or $H\underline{q} = -X\underline{p}$)

iff $E(c+\underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ & $H\underline{q} \in C(X)$

Note: Under Model III if H is invertible, we would use $\underline{\lambda}'(X'H^{-1}X)^{-}X'H^{-1}\underline{y}$ to estimate $\underline{\lambda}'\underline{\beta}$ & this is unbiased but also $H\underline{q} = HH^{-1}X(X'H^{-1}X)^{-}\underline{\lambda}$ or $H\underline{q} = X(X'H^{-1}X)^{-}\underline{\lambda} \in C(X)$

Remark: Corollary 1 implies that $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $H\underline{a} \in \mathcal{C}(X)$.

$E(\underline{a}'\underline{y}) = \underline{g}'X\underline{\beta}$ is unbiased for $\underline{g}'X\underline{\beta}$

So, $\underline{a}'\underline{y}$ is BLUE iff $H\underline{a} \in C(X)$

Corollary 2. Under Model III, if $c+\underline{a}'\underline{y}$ and $k+\underline{b}'\underline{y}$ are both BLUEs of the same estimable function, then $\text{Var}[c+\underline{a}'\underline{y}-(k+\underline{b}'\underline{y})] = 0$ or equivalently $c+\underline{a}'\underline{y} = k+\underline{b}'\underline{y}$ with probability 1.

BLUEs are unique wp 1

Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$.

For estimable $\underline{\lambda}'\underline{\beta}$ ($\underline{\lambda}' \in \mathcal{R}(X)$), a solution to equations (1) is given by

$$\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G_{12}\underline{\lambda} \\ G_{22}\underline{\lambda} \end{bmatrix}.$$

Because $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$ is symmetric, G' is also a generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$.

Another solution to the equations (1) is then given by

$$\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}' \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G'_{11} & G'_{21} \\ G'_{12} & G'_{22} \end{bmatrix} \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G'_{21}\underline{\lambda} \\ G'_{22}\underline{\lambda} \end{bmatrix}.$$

Thus, under Model III, BLUEs of an estimable function $\underline{\lambda}'\underline{\beta}$ are given by $\underline{\lambda}'G'_{12}\underline{y}$ and $\underline{\lambda}'G_{21}\underline{y}$. By Corollary 2 to Theorem 2.20, $\underline{\lambda}'G'_{12}\underline{y} = \underline{\lambda}'G_{21}\underline{y}$ with probability 1.

Let $\Lambda'_1\underline{\beta}$ and $\Lambda'_2\underline{\beta}$ be estimable vectors. Then, the elements of $\Lambda'_1G'_{12}\underline{y}$ and $\Lambda'_2G'_{12}\underline{y}$ are BLUEs of the elements of $\Lambda'_1\underline{\beta}$ and $\Lambda'_2\underline{\beta}$, respectively.

If we replace $X'X$ with H and A' with X in Theorem 2.14 and Lemma 2.13, then these results hold for $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$, any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$, and can be used to show that,

$$\text{cov}[\Lambda'_1G'_{12}\underline{y}, \Lambda'_2G'_{12}\underline{y}] = \sigma^2\Lambda'_1(-G_{22})\Lambda_2,$$

and an unbiased estimator of σ^2 is given by

$$\widehat{\sigma}^2 = \underline{y}'G_{11}\underline{y}/\left[\text{rank}(H, X) - \text{rank}(X)\right].$$

Model IIIa, where q constraints $A\underline{\beta} = \underline{b}$ are imposed on the parameter space, can be approached as a special case of Model III.

The vector $\underline{b}$ can be regarded as a vector of q additional "data points," in which case the vector of all data points is given by $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$.

We have

$$\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix} = \begin{bmatrix} X \\ A \end{bmatrix} \underline{\beta} + \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix}$$

and

$$E \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \underline{0} \quad \text{and} \quad \text{Var} \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \sigma^2 \begin{bmatrix} H & 0 \\ 0 & 0 \end{bmatrix}.$$

If we take the parameter space to be $\{\underline{\beta}, \sigma^2 : \underline{\beta} \in R^p, \sigma^2 > 0\}$, we arrive at an unconstrained model for $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$ having the form of Model III.

It can be verified that BLUEs of any estimable function $\underline{\lambda}'\underline{\beta}$ computed on the basis of this model are BLUEs of $\underline{\lambda}'\underline{\beta}$ under the original constrained model, and vice versa.

2.11 When Is the Simple Least Squares Estimator a Best Linear Unbiased Estimator?

(simple/ordinary LS ≡ SLSE)

Take the model to be Model III, and suppose that $\underline{\lambda}'\underline{\beta}$ is an estimable function. The simple (or ordinary) least squares estimator (SLSE) of $\underline{\lambda}'\underline{\beta}$ is given by $\underline{\lambda}'\widetilde{\underline{\beta}}$, where $\widetilde{\underline{\beta}}$ is any solution to $X'X\widetilde{\underline{\beta}} = X'\underline{y}$. The SLSE is an unbiased estimator, but in general, it will not be a best linear unbiased estimator (BLUE).

$\widetilde{\beta} = (X'X)^{-}X'y$

i.e. for estimable $\lambda'\beta$, $\lambda'(X'X)^{-}X'y$ is SLSE

Under what circumstances is the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ a BLUE? An answer is given by the following theorem.

Theorem 2.21. *Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that*

$$HX\underline{r} = X\underline{q},$$

where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.

$\hat{\beta}$ sol to NEs

proof: the SLSE of $\lambda'\beta$ is given by $\underline{r}'X'\underline{y} = \lambda'\hat{\beta}$

$\underline{r}$ is solution to conjugate NEs $X'X\underline{r} = \lambda$

This is unbiased since $E(\underline{r}'X'\underline{y}) = \underline{r}'X'X\beta = \lambda'\beta$

By Remark/Theorem 2.20, $\underline{r}'X'\underline{y}$ is BLUE iff $HX\underline{r} \in \mathcal{C}(X)$

Under what circumstances is the SLSE of every estimable function $\underline{\lambda}'\underline{\beta}$ a BLUE? An answer is provided by the following corollary.

iff $HX\underline{r} = X\underline{q}$ for some $\underline{q}$

Corollary 1. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a matrix Q such that

$$HX = XQ.$$

Note: $H = Var(\underline{\epsilon})$ is $n \times n$
& X is $n \times p$
& Q is $p \times p$

Proof. Note that $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X\underline{r} = \underline{\lambda}$ for some $\underline{r}$ (by Theorem 2.7). Thus, by Theorem 2.21, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if , for every $\underline{r}$, there exists a $\underline{q}$ such that

$$HX\underline{r} = X\underline{q}.$$

Need to show: for any $\underline{r} \in \mathbb{R}^p$, $HX\underline{r} = X\underline{q}$ holds for some $\underline{q}$ (depending on $\underline{r}$)

$\Leftrightarrow$ $HX = XQ$ for some $p \times p$ matrix Q

$\Rightarrow$ let $\underline{r}_i$ be the ith column of $p \times p$ identity matrix I_p

& $HX\underline{r}_i = X\underline{q}_i$ for some $\underline{q}_i$, $i = 1, \ldots, p$.

Then, $HX = HXI_p = HX[\underline{r}_1, \ldots \underline{r}_p] = X[\underline{q}_1, \ldots, \underline{q}_p]$

$\Leftarrow$ Suppose $HX = XQ \Rightarrow HX\underline{r} = XQ\underline{r} = X\underline{q} = XQ$, $Q = [\underline{q}_1 \ldots \underline{q}_p]$

The next corollary gives a condition $HP_X = P_XH$ that is equivalent to $HX = XQ$ for some Q given in Corollary 1, where $P_X = X(X'X)^-X'$. Note that $(HP_X)' = P_X'H' = P_XH$ is not necessarily a symmetric matrix.

holds for any $\underline{r} \in \mathbb{R}^p$

$P_X \equiv X(X'X)^-X'$ projection matrix of X Note: $P_X' = P_X$, $H' = H$

Corollary 2. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $HP_X = P_XH$ (that is, HP_X is a symmetric matrix).

$HP_X = P_XH$ holds $\Leftrightarrow HX = XQ$

$= (HP_X)'$

Proof. By Corollary 1, it suffices to show that $HP_X = P_XH$ (that is, HP_X is a symmetric matrix) if and only if $HX = XQ$ for some Q.

$\Rightarrow$ Suppose $HP_X = P_XH$, then multiply both sides by X

get $HP_XX = P_XHX$ or $HX = XQ$ for $Q = (X'X)^-X'HX$

HP_X X P_X $X(X'X)^-X'$ X So SLSE must be BLUE. (corollary 1)

$\Leftarrow$ If $HX = XQ$, then

$P_XHP_X = P_XHX(X'X)^-X' = P_XXQ(X'X)^-X' = XQ(X'X)^-X'$

symmetric since $P_X' = P_X$, $H' = X$ $= HX(X'X)^-X' = HP_X$

So, HP_X is symmetric i.e. $HP_X = (HP_X)' = P_X H$

Remarks. Theorem 2.21 and its two corollaries are still valid if $\text{Var}(\underline{y}) = \sigma^2 H$ is put in place of H.

Furthermore, under Model VII with variance V:

1. Theorem 2.21 and its two corollaries are still valid if V is put in place of H.

2. That is, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that $VX\underline{r} = X\underline{q}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$;

3. And, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a Q such that $VX = XQ$ or if and only if $VP_X = P_X V$.

4. Above, elements of $\underline{q}$ and Q may be functions of $\underline{\theta}$.

5. For more details, see George Zyskind's paper, "On canonical forms, non-negative covariance matrices and best and simple least squares linear estimators in linear models" (*The Annals of Mathematical Statistics*, Volume 38, 1967, 1092–1109).